



Class Project Phase 2

Group: 1

Team Members:

John Bostater

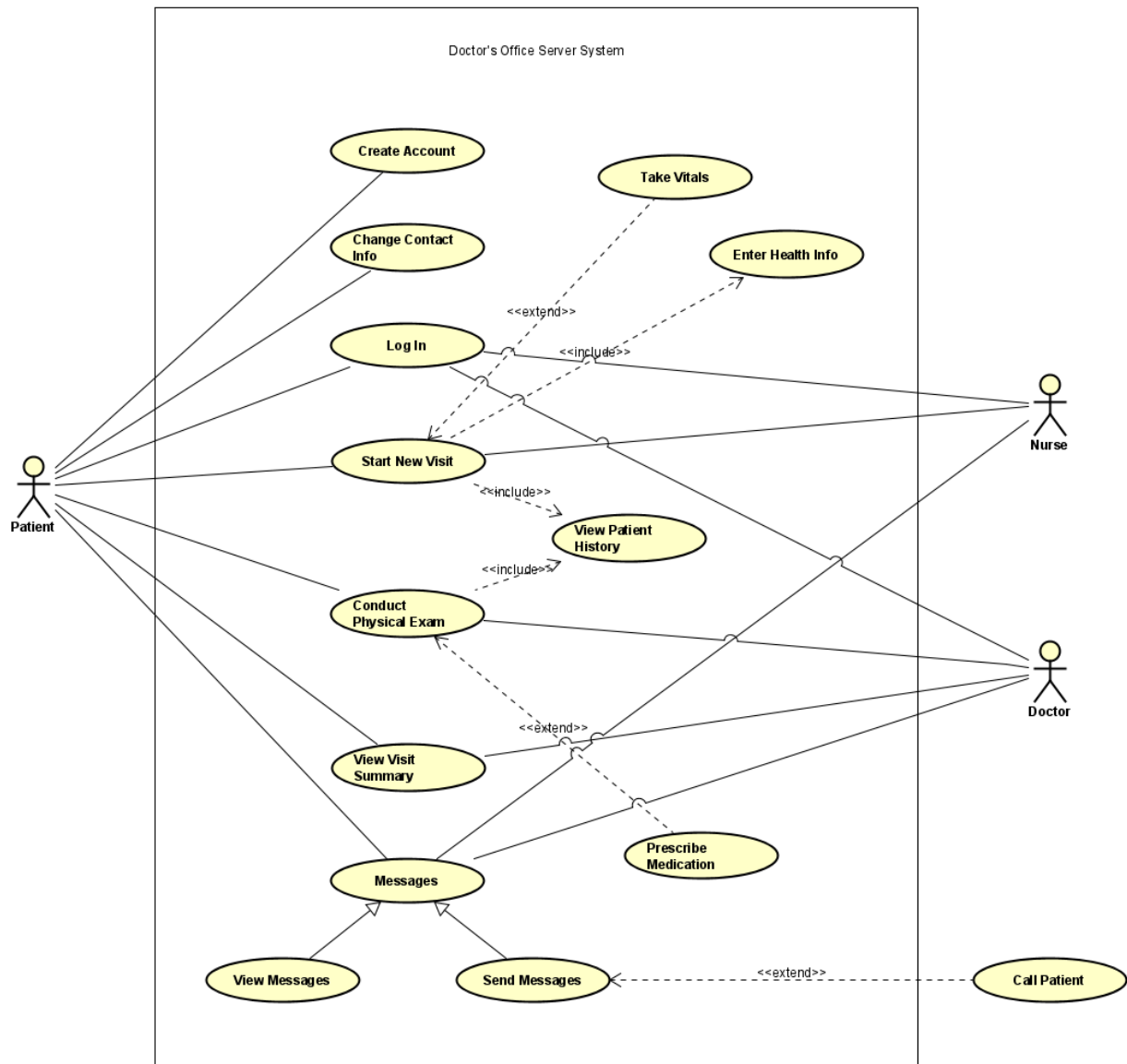
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1. Use Cases and the Use Case Diagram (30). Copy and paste your use case diagram from Astah



Create Account	
Actors	Patient
Description	Patient creates an account in the system using their personal information. Once the account is created, the patient can log in to view visit summaries and message the nurse/doctor.
Data	Patient's first name, last name, and birthday.
Stimulus	Patient clicking Create Account button in the Patient Portal login screen.
Response	Open the Account Creation page for the patient to input their information and press "Confirm".
Comments	Patient accounts and all associated info will be stored in the database.

Change Contact Info	
Actors	Patient
Description	Patient views the contact info on file with the ability to modify the data with new information. The patient can then confirm the updated information to update the database with the new information.
Data	Patient's phone number, insurance provider, and pharmacy provider.
Stimulus	Patient clicking Change Patient Information button in Patient Portal menu.
Response	Open the Patient Information page for the patient to update their information and press "Update Information" to confirm the changes.
Comments	Updated information overwrites old

	information stored in the database.
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Log In	
Actors	Patient, Nurse, Doctor
Description	Nurses and doctors log in to the Staff Portal page, and patients log in to the Patient Portal page with their account information.
Data	Patient's first name, last name, and birthday (for Patient log in).
Stimulus	Patient clicking "Patient" in the Login start page. Nurses and Doctors clicking "Staff" in the Login start page, and then their respective button on the Staff Login page.
Response	Patients log into the menu of the Patient Portal, where they can access all of their information. Nurses/doctors log into their respective staff portal menus.
Comments	

Start New Visit	
Actors	Patient, Nurse
Description	Nurses open a new Visit form for the patient with their personal info.
Data	Patient's first name, last name, and birthday.
Stimulus	Nurse clicks "New Visit Form" in the menu of the Nurse Portal.
Response	Open the New Visit form page and the ability to input all relevant data, then submit form to doctor by clicking "Submit

	for Physical.”
Comments	Nurse will enter the patient's health info into the form. If the patient is over 12, the nurse will also take the patient's vitals and input them into the form. Completed visit form will be delivered to the Doctor to perform physical exam and then the summary will be stored in the database for patient and doctor to review.

Take Vitals	
Actors	Patient, Nurse
Description	If the patient is over 12, the nurse will take the patient's vitals and input them into the form.
Data	Patient's weight, height, body temperature, and blood pressure
Stimulus	Nurse fills out the patient's personal information. If the patient's date of birth indicates they are over 12, the option to enter the patient's vitals is prompted.
Response	Nurse fills out the vitals on the visit form along with other data, then clicks “Submit for Physical.”
Comments	Only performed if the patient is over 12.

Enter Health Info	
Actors	Patient, Nurse
Description	Nurse asks patient relevant health questions and inputs responses into the visit form as part of starting new visit.
Data	Patient's known allergies and other health

	concerns.
Stimulus	Nurse clicks “New Visit Form” in the menu of the Nurse Portal and has filled out the patient’s personal information.
Response	Nurse fills out the health info for the patient along with other data, then clicks “Submit for Physical.”
Comments	Health info is added to the visit summary text file in the database.

View Patient History	
Actors	Nurse, Doctor
Description	Nurse and doctor can view a patient’s medical history when entering or reviewing the visit form.
Data	Patient’s previously prescribed medications, immunization history, and previous health issues.
Stimulus	Nurse clicks “New Visit Form” in the menu of the Nurse Portal and has filled out the patient’s personal information. Doctor clicks “Patient Visit Form,” opening a form the nurse has submitted.
Response	Patient’s previous history is loaded from a file.
Comments	Patient’s previous history is stored in a text file in the database.

Conduct Physical Exam	
Actors	Patient, Doctor
Description	Doctor reviews the visit form submitted by

	the nurse, then conducts the physical exam with the patient and inputs the results.
Data	Patient's physical exam results and other notes.
Stimulus	Doctor clicks "Patient Visit Form" in the menu of the Doctor Portal, then "Conduct Exam" after reviewing the form.
Response	Doctor fills out the physical exam results, other notes, then clicks "Update Summary."
Comments	Results and other notes are added to the visit summary text file in the database.

Prescribe Medication	
Actors	Doctor
Description	If during the physical exam the doctor has deemed it necessary, they prescribe the patient medication by filling out the script, signing it, and sending it to the pharmacy.
Data	Patient's prescription script, doctor's signature
Stimulus	Doctor clicks "Conduct Exam" after reviewing the visit form and has deemed it necessary to prescribe the patient medication.
Response	Doctor fills out the prescription script and provides their signature, then clicks "Send Prescription."
Comments	Prescription is sent to the patient's pharmacy as loaded from the relevant text file in the database.

View Visit Summary	
Actors	Patient, Doctor
Description	Patient and doctor can review a summary of previous visits.
Data	For patient: Visit date For doctor: Select patient from dropdown, visit date
Stimulus	Patient clicks “View Previous Visits” in the menu of the Patient Portal. Doctor clicks “View Patient Records” in the menu of the Doctor Portal then selects a patient from the dropdown.
Response	Previous visit is loaded from text file when a date is selected.
Comments	Visit summaries are stored as text files in the database.

Messages	
Actors	Patient, Nurse, Doctor
Description	Patient and the nurse/doctor can view and send messages back and forth.
Data	See View Messages and Send Messages
Stimulus	User clicks “Messages” from their respective portal menu.
Response	See View Messages and Send Messages
Comments	Messages is split into viewing and sending messages. Messages are stored in a text file in the database.

View Messages

Actors	Patient, Nurse, Doctor
Description	Patient can view messages from the office, and nurse/doctor can view messages from the patient.
Data	For patient: None For nurse/doctor: Select patient from dropdown
Stimulus	See Messages
Response	Load messages between the patient and the nurse/doctor.
Comments	Messages are stored in a text file in the database.

Send Messages	
Actors	Patient, Nurse, Doctor
Description	Patient can send messages to the office, and nurse/doctor can respond.
Data	Message to be sent
Stimulus	See Messages
Response	User types out their message, then clicks "Send" to deliver it.
Comments	Sent messages are appended to the messages text file in the database.

Call Patient	
Actors	Nurse, Doctor
Description	If there is an emergency, the nurse/doctor can call the patient using the phone number on record.

Data	Patient's phone number
Stimulus	Nurse/doctor calls the phone number loaded from the text file in the Staff Messages for the patient.
Response	n/a
Comments	Calling patient is done outside of the system. Phone number is stored with patient's other information in a file in the database.

2. Vc Object Identification and CRC (20). List each object and CRC for each object.

Class: System

The system database that stores patient and appointment histories and handles many of the operations end-users perform, such as sending messages, conducting exams, storing information, and handling calls.

Class: Appointment

A physical exam detailing the patient's vitals, credentials, health history, and health concerns taken during screening, as well as the results of the actual exam and any prescription changes after the doctor examines the patient. The system uses these to curate patients' appointment histories and to save appointment dates.

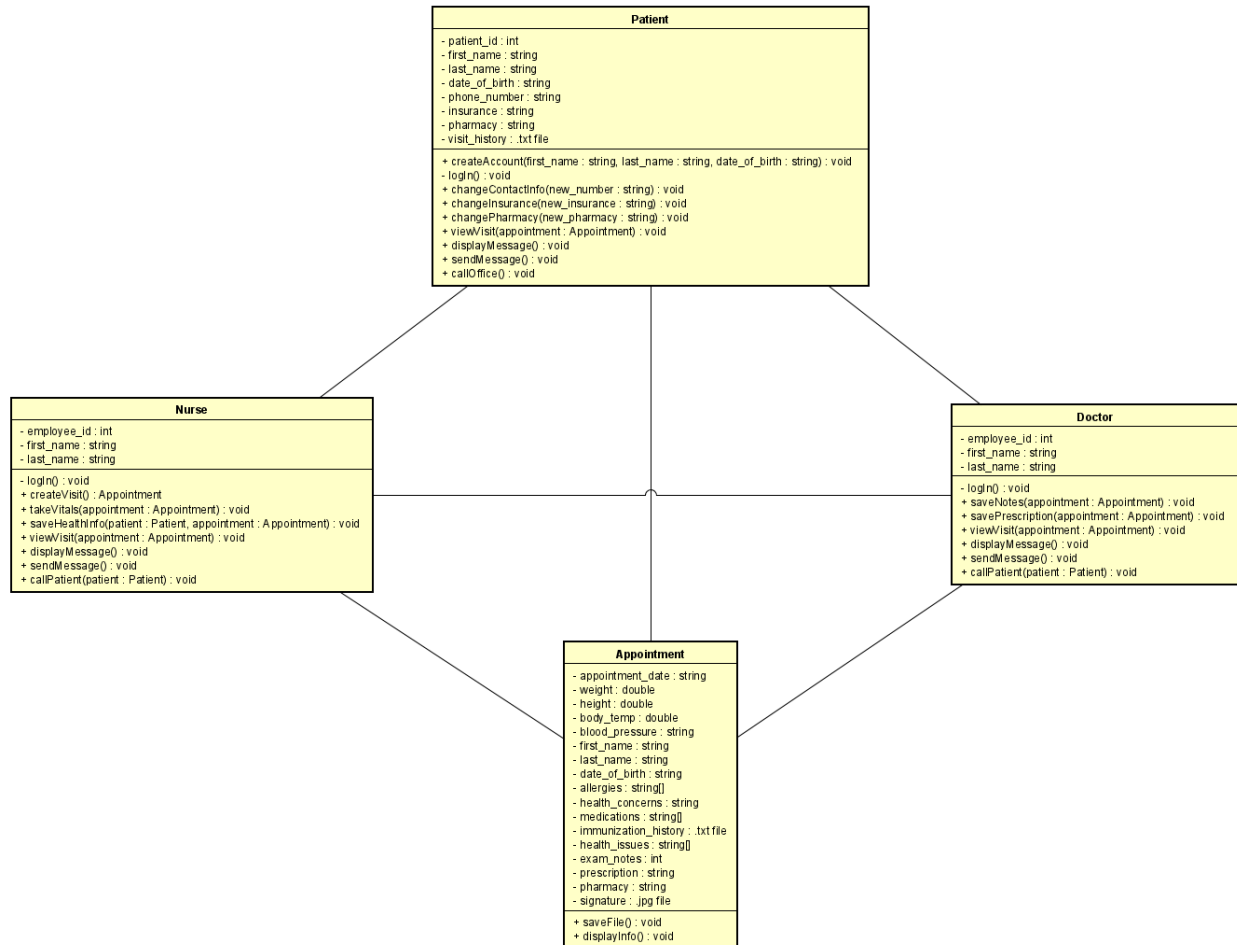
Class: Patient	Uses the system to view their exam results and history. They register themselves into the system, attend physical exams, and discuss prescription plans with doctors and nurses.
Responsibility:	Collaborator:
Create Account	System
Change Contact Info	System
Log In	System
View Visit Summary	System, Appointment
View Messages	System
Send Messages	System, Nurse, Doctor
Call Office (Call Patient)	System

Class: Nurse	Takes the patient's vitals, current information, and notes any key information such as allergens and new health concerns. They use the system to submit this information for a physical exam (appointment) and can converse with patients via the system's messaging forum.
Responsibility:	Collaborator:
Log In	System
Start New Visit	System, Patient
Take Vitals	System, Patient
Enter Health Info	System, Patient
View Patient History	System, Appointment
View Messages	System
Send Messages	System, Patient, Doctor
Call Patient	System, Patient

Class: Doctor	Conducts physical exams and evaluates the exam results as well as the patient's history to determine their best course of treatment. They use the system to view patient history/nurse's notes and assign prescriptions to patients. Like nurses, they also converse with patients to discuss treatment plans through the messaging forum.
Responsibility:	Collaborator:
Log In	System
Conduct Physical Exam	System, Patient
View Patient History	System, Appointment

Prescribe Medication	System
View Visit Summary	System, Appointment
View Messages	System
Send Messages	System, Patient, Nurse
Call Patient	System, Patient

3. Class Diagram (40). Copy and paste your class diagram from Astah. Class in the class diagram need to have attributes, methods and classes need to be connected using appropriate UML class relationships.



4. Test Plan for Functional Testing (30)

Scenario 1: A 15 yr old teen visits for an appointment.

Scenario 2: Doctor conducts a physical exam for a patient.

Scenario 3: A patient receives a message from the nurse.

Scenario 4: A patient wants to contact a staff member about their latest visit.

Scenario 5: Doctor notices something about a patient and decides to call them.

Case	Scenario/Inputs	Expected Output
1	1. Nurse clicks “New Visit Form” via the Nurse Portal. 2. Nurse takes Patient’s vitals. 3. Nurse fills out the rest of the form. - Patient’s first name - Patient’s last name - Patient’s birthday	1. A blank Visit Form should open. 2. Results should be added to form. 3. Form is stored in database and sent to Doctor.
2	1. Doctor clicks “Patient Visit Form” via the Doctor Portal. 2. Doctor clicks on “Conduct Exam.” 3. Doctor saves exam results after filling out the form. - Exam results - Notes	1. Patient’s form should appear. 2. Patient’s summary should open. 3. Old summary in database is replaced with the recent one.
3	1. Patient clicks on “View Messages” in the Messages window. 2. Patient clicks on Nurse’s message.	1. Patient should be sent to a screen that shows messages. 2. Patient should be shown Nurse’s entire message.
4	1. Patient clicks on “Send Message” in the Messages window. 2. Patient clicks “Send” to send message to Doctor/Nurse.	1. A text box appears for the patient to enter their message. 2. Message is sent to Doctor/Nurse and should be in the database.
5	1. Doctor clicks on Patient’s file. 2. Doctor calls Patient via the phone number on file.	1. Patient’s information should appear to Doctor. 2. Patient should receive the phone call from the Doctor.

Credit Sheet

Team Member Name	Contributions
John Bostater	Prototype Implementation
Tristan Andrade	Object ID and CRC Diagram
Austin Mayhew	Use Case Diagram
Ryan Clark	Class Diagram
Juan Rosas Jr.	Test Plan